

◆ Gemini

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# AIM:- Morphological operation using Black hat technique.
# PROGRAM:-
import cv2
import numpy as np # Import numpy for array operations
import matplotlib.pyplot as plt # Import matplotlib for displaying images

# Correct image path to one available in Colab environment
image_path = "V.KOHLI.jpeg" # Using V.KOHLI.jpeg as it is available in the kernel

try:
    ... input_image = cv2.imread(image_path) # Read the image in color

    ... # Check if image was loaded successfully
    ... if input_image is None:
    ...     raise FileNotFoundError(f"Error: Image not found at {image_path}. Please ensure it's uploaded")

    ... filterSize = (3, 3)
    ... kernel = cv2.getStructuringElement(cv2.MORPH_RECT, filterSize)

    ... # Convert the input image to grayscale for morphological operations as Black Hat typically works
    ... gray_image = cv2.cvtColor(input_image, cv2.COLOR_BGR2GRAY)

    ... blackhat_img = cv2.morphologyEx(gray_image, cv2.MORPH_BLACKHAT, kernel)

    ... # Display the original (color) and Black Hat images using matplotlib
    ... plt.figure(figsize=(10, 5))

    ... plt.subplot(1, 2, 1)
    ... plt.imshow(cv2.cvtColor(input_image, cv2.COLOR_BGR2RGB)) # Display original color image
    ... plt.title('Original Image (Color)')
    ... plt.axis('off')

    ... plt.subplot(1, 2, 2)
    ... plt.imshow(blackhat_img, cmap='gray') # Display Black Hat image in grayscale
    ... plt.title('Black Hat Image (Grayscale)')
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plt.title('Black Hat Image (Grayscale)')
... plt.axis('off')

... plt.show()

... # Save the Black Hat image
... cv2.imwrite("blackhat.jpg", blackhat_img)
... print("Black Hat image saved as 'blackhat.jpg'")

except FileNotFoundError as e:
... print(e)
except Exception as e:
... print(f"An unexpected error occurred: {e}")

# cv2.imshow and cv2.waitKey do not work in Google Colab
# cv2.destroyAllWindows() also not needed in Colab for display
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Original Image (Color)



Black Hat Image (Grayscale)



Black Hat image saved as 'blackhat.jpg'